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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
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Roll Number	2520030138
Branch	CSE
Course Outcome	5

Case Study Topic: UrbanAI – Intelligent Smart City Infrastructure Optimization Platform

Work Done Summary:

In this case study, the Quick Sort algorithm was implemented to manage the scheduling priority within an automated Smart City Resource Allocation System. Grid infrastructure load values and energy consumption demands were stored in an array and sorted in descending order using a high-efficiency pivot strategy. The platform automatically organizes and ranks city sectors based on real-time load telemetry, enabling immediate identification of heavily strained zones requiring immediate resource deployment. Quick Sort proved highly capable of processing massive datasets generated by citywide IoT sensor grids, offering near-instant ranking refreshes. The implementation presents a highly practical mechanism for algorithmic load-balancing and grid maintenance scheduling within smart city architectures.

Implementation Details:

- Stored infrastructure load and consumption metrics in an un-sorted integer array.
- Implemented the Quick Sort framework leveraging a pivot-driven partitioning mechanism.
- Selected the terminal array element as the default pivot for partitioning.
- Sorted all monitored grid metrics in descending order to isolate high-priority targets.
- Generated structural maintenance priority rankings immediately post-sorting.
- Displayed system load profiles explicitly before and after sorting operations.
- Successfully isolated and flagged the highest-load sector requiring immediate load-shedding or support.
- Evaluated sorting speed and performance using programmatic complexity benchmarks.

Result: Screenshots / Output

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============================================================
 URBANAI – INFRASTRUCTURE RESOURCE ALLOCATION SYSTEM
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Grid Sector Loads Before Sorting (kW):
890 1200 760 1450 980 1100 1670 1340

Priority Allocations (Highest Load to Lowest):
1670 1450 1340 1200 1100 980 890 760

Peak Monitored Load Sector: 1670 kW

Top 3 High-Strain Sectors:
Rank 1 : 1670 kW
Rank 2 : 1450 kW
Rank 3 : 1340 kW

Complexity Analysis:
Best Case    : O(n log n)
Average Case : O(n log n)
Worst Case   : O(n²)

Resource Allocation Schedule Generated Successfully.
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Time Complexity:

- Best Case : $O(n \log n)$
- Average Case : $O(n \log n)$
- Worst Case : $O(n^2)$

Conclusion:

The UrbanAI Infrastructure Resource Allocation System successfully organized and categorized metropolitan sector load priorities using the Quick Sort methodology. By deploying a pivot-based divide-and-conquer strategy, the platform efficiently processes massive input arrays to output real-time dispatch rankings. This case study reinforces how fundamental sorting logic directly supports large-scale municipal logistics and automated grid management platforms.

GitHub Link:

https://github.com/marahrudaydeepak-ctrl/DSA-2

Student Signature	**Faculty Signature**
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